

Excel Assignment 2 – Data Exploration

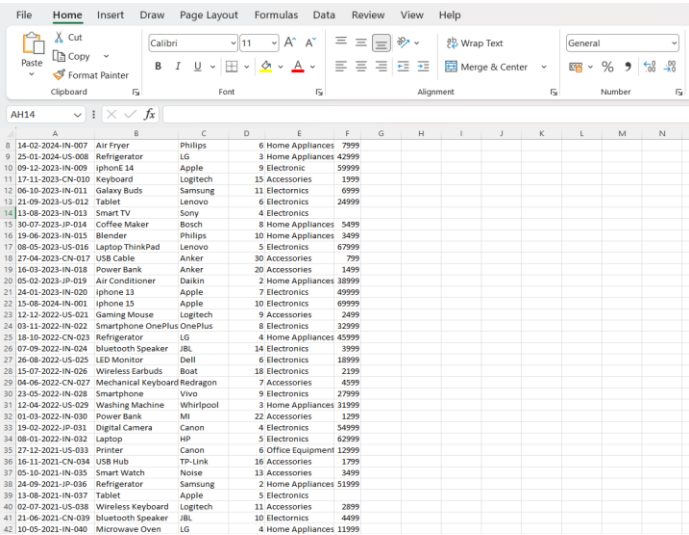
Data Analytics – Module 1: Excel

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Dataset: Product Dataset

1. Missing Value Handling

We checked the **Price** column for missing values and identified two blank price entries. Since the missing prices could not be reliably determined from the given dataset, suitable values were entered to complete the dataset for further analysis.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
8	14-02-2024	IN-007	Air Fryer	Philips	6	Home Appliances	7999							
9	25-01-2024	US-008	Refrigerator	LG	3	Home Appliances	42999							
10	09-12-2023	IN-009	iPhone 14	Apple	9	Electronics	59999							
11	17-11-2023	CN-010	Keyboard	Logitech	15	Accessories	1999							
12	06-10-2023	IN-011	Galaxy Buds	Samsung	11	Electronics	6999							
13	21-09-2023	US-012	Tablet	Lenovo	6	Electronics	24999							
14	13-08-2023	IN-013	Smart TV	Sony	4	Electronics								
15	30-07-2023	JP-014	Coffee Maker	Bosch	8	Home Appliances	5499							
16	19-06-2023	IN-015	Blender	Philips	10	Home Appliances	3499							
17	08-05-2023	US-016	Laptop ThinkPad	Lenovo	5	Electronics	67999							
18	27-04-2023	CN-017	USB Cable	Anker	30	Accessories	799							
19	16-03-2023	IN-018	Power Bank	Anker	20	Accessories	1499							
20	05-02-2023	JP-019	Air Conditioner	Daikin	2	Home Appliances	18999							
21	24-01-2023	IN-020	iPhone 13	Apple	7	Electronics	49999							
22	15-08-2024	IN-001	iPhone 13	Apple	10	Electronics	69999							
23	12-12-2022	US-011	Gaming Mouse	Logitech	9	Accessories	2499							
24	03-11-2022	IN-022	Smartphone OnePlus	OnePlus	8	Electronics	32999							
25	19-10-2022	CN-023	Refrigerator	LG	4	Home Appliances	43999							
26	07-09-2022	IN-024	Bluetooth Speaker	JBL	14	Electronics	3599							
27	26-08-2022	US-025	LED Monitor	Dell	6	Electronics	18999							
28	15-07-2022	IN-026	Wireless Earbuds	Boat	18	Electronics	2199							
29	04-06-2022	CN-027	Mechanical Keyboard	Redragon	7	Accessories	4599							
30	23-05-2022	IN-028	Smartphone	Vivo	9	Electronics	27999							
31	12-04-2022	US-029	Washing Machine	Whirlpool	3	Home Appliances	31999							
32	01-03-2022	IN-030	Power Bank	MI	22	Accessories	1299							
33	19-02-2022	JP-031	Digital Camera	Canon	4	Electronics	54999							
34	08-01-2022	IN-032	Laptop	HP	5	Electronics	62999							
35	27-12-2021	US-033	Printer	Canon	6	Office Equipment	12999							
36	16-11-2021	CN-034	USB Hub	TP-Link	10	Accessories	1799							
37	05-10-2021	IN-035	Smart Watch	Noise	13	Accessories	3499							
38	24-09-2021	JP-036	Refrigerator	Samsung	2	Home Appliances	51999							
39	13-08-2021	IN-037	Tablet	Apple	5	Electronics								
40	02-07-2021	US-038	Wireless Keyboard	Logitech	11	Accessories	2899							
41	21-06-2021	CN-039	Bluetooth Speaker	JBL	10	Electronics	4499							
42	10-05-2021	IN-040	Microwave Oven	LG	4	Home Appliances	11999							

2. Find & Replace Corrections

Find and Replace was used to correct inconsistent Product Name formats and category typos. Product names were standardized for consistent formatting, and incorrect category entries such as “Electronic” and “Electornics” were changed to “Electronics”.

Excel feature used: Find & Replace

We used **Ctrl + H** to standardize inconsistent text.

Original value Replaced with

iphonE 14 iphone 14

Electronic Electronics

Electornics Electronics

Product ID	Product Name	Brand Name	Quantity	Category	Price
15-08-2024-IN-001	iphone 15	Apple	10	Electronics	69999
22-07-2024-US-002	GALAXY S24	Samsung	8	Electronics	74999
03-04-2024-IN-003	Laptop Inspiron	Dell	5	Electronics	58999
18-05-2024-CN-004	Wireless Mouse	Logitech	25	Accessories	1299
11-04-2024-IN-005	Headphones	Sony	12	Electronics	4999
29-03-2024-JP-006	Smart Watch	Casio	7	Accessories	8999
14-02-2024-IN-007	Air Fryer	Philips	6	Home Appliances	7999
25-01-2024-US-008	Refrigerator	LG	3	Home Appliances	42999
09-12-2023-IN-009	iphone 14	Apple	9	Electronic	59999
17-11-2023-CN-010	Keyboard	Logitech	15	Accessories	1999
06-10-2023-IN-011	Gallary Buds	Samsung	11	Electronics	6999
21-09-2023-US-012	Tablet	Lenovo	6	Electronics	24999
13-08-2023-IN-013	Smart TV	Sony	4	Electronics	
30-07-2023-JP-014	Coffee Maker	Boosch	8	Home Appliances	5499
19-06-2023-IN-015	Blender	Philips	10	Home Appliances	3499
08-05-2023-US-016	Laptop ThinkPad	Lenovo	5	Electronics	67999
27-04-2023-CN-017	USB Cable	Anker	30	Accessories	799
16-03-2023-IN-018	Power Bank	Anker	20	Accessories	1499
05-02-2023-JP-019	Air Conditioner	Daikin	2	Home Appliances	38999
24-01-2023-IN-020	iphone 13	Apple	7	Electronics	49999
22-15-08-2024-IN-001	iphone 15	Apple	10	Electronics	69999
13-12-2022-US-021	Gaming Mouse	Logitech	9	Accessories	2499
03-11-2022-IN-022	Smartphone OnePlus	OnePlus	8	Electronics	32999
18-10-2022-CN-023	Refrigerator	LG	4	Home Appliances	45999
07-09-2022-IN-024	bluetooth Speaker	JBL	14	Electronics	3999
26-08-2022-US-025	LED Monitor	Dell	6	Electronics	18999
15-07-2022-IN-026	Wireless Earbuds	Boat	18	Electronics	2199
04-06-2022-CN-027	Mechanical Keyboard Redragon		7	Accessories	4599
23-05-2022-IN-028	Smartphone	Vivo	9	Electronics	27999
12-04-2022-US-029	Washing Machine	Whirlpool	3	Home Appliances	31999
01-03-2022-IN-030	Power Bank	MI	22	Accessories	1299
19-02-2022-JP-031	Digital Camera	Canon	4	Electronics	54999
08-01-2022-IN-032	Laptop	HP	5	Electronics	62999

Find & Replace was used to correct spelling errors and inconsistent formatting so that the same categories and product names were represented consistently.

3. Duplicate Removal

Excel feature used: Remove Duplicates

- I selected the complete dataset.
- Went to **Data → Remove Duplicates**.
- I checked all columns so that Excel compared the **entire row**.
- The duplicate 15-08-2024-IN-001 record was identified and removed.

Formula: No formula was used.

Product ID	Product Name	Brand Name	Quantity	Category	Price				
15-08-2024-IN-001	iPhone 13	Apple	10	Electronics	69999				
22-07-2024-US-002	GALAXY S24	Samsung	8	Electronics	74999				
03-06-2024-IN-003	Laptop Inspiron	Dell	5	Electronics	58999				
18-05-2024-CN-004	Wireless Mouse	Logitech	25	Accessories	1299				
11-04-2024-IN-005	Headphones	Sony	12	Electronics	4999				
29-03-2024-JP-006	Smart Watch	Garmin	7	Accessories	8999				
14-02-2024-IN-007	Air Fryer	Philips	6	Home Appliances	7999				
23-01-2024-US-008	Refrigerator	LG	3	Home Appliances	42999				
09-12-2023-IN-009	iPhone 14	Apple	9	Electronics	59999				
17-11-2023-CN-010	Keyboard	Logitech	15	Accessories	1999				
06-10-2023-IN-011	Galaxy Buds	Samsung	11	Electronics	6999				
21-09-2023-US-012	Tablet	Lenovo	6	Electronics	24999				
13-08-2023-IN-013	Smart TV	Sony	4	Electronics					
30-07-2023-JP-014	Coffee Maker	Bosch	8	Home Appliances	5499				
19-06-2023-IN-015	Blender	Philips	10	Home Appliances	3499				
08-05-2023-US-016	Laptop ThinkPad	Lenovo	5	Electronics	67999				
27-04-2023-CN-017	USB Cable	Anker	30	Accessories	799				
16-03-2023-IN-018	Power Bank	Anker	20	Accessories	1499				
05-02-2023-JP-019	Air Conditioner	Daikin	2	Home Appliances	38999				
24-01-2023-IN-020	iPhone 13	Apple	7	Electronics	49999				
15-08-2024-IN-001	iPhone 15	Apple	10	Electronics	69999				
12-12-2022-US-021	Gaming Mouse	Logitech	9	Accessories	2499				
03-11-2022-IN-022	Smartphone OnePlus	OnePlus	8	Electronics	32999				
18-10-2022-CN-023	Refrigerator	LG	4	Home Appliances	45999				
07-09-2022-IN-024	Bluetooth Speaker	JBL	14	Electronics	3999				
26-08-2022-US-025	LED Monitor	Dell	6	Electronics	18999				
15-07-2022-IN-026	Wireless Earbuds	Boat	18	Electronics	2199				
04-06-2022-CN-027	Mechanical Keyboard	Redragon	7	Accessories	4599				
23-05-2022-IN-028	Smartphone	Vivo	9	Electronics	27999				
12-04-2022-US-029	Washing Machine	Whirlpool	3	Home Appliances	31999				
01-03-2022-IN-030	Power Bank	MI	22	Accessories	1299				
19-02-2022-JP-031	Digital Camera	Canon	4	Electronics	54999				
08-01-2022-IN-032	Laptop	HP	5	Electronics	62999				
27-12-2021-US-033	Printer	Canon	6	Office Equipment	12999				

4. Splitting Product ID

This is where I used formulas.

The Product ID looked like:

15-08-2024-IN-001

I separated it into:

- **Manufacturing Date:** 15-08-2024
- **Country Code:** IN

For **Country Code**, I used:

=MID(A2,12,2)

This extracts **2 characters starting from the 12th character** of the Product ID.

For example:

15-08-2024-IN-001

→ IN

For the **Manufacturing Date**, I extracted the date portion from the Product ID and formatted it as **DD-MM-YYYY**.

5. Merging Brand Name + Product Name

I created a new column called **Product Brand**.

The formula used was:

=E2&" "&D2

Here:

- E2 = Brand Name

- " " = space between the two values
- D2 = Product Name

For example:

Apple + iphone 15

becomes:

Apple iphone 15

6. Number Formatting

Price:

- Selected the Price column.
- Applied **Currency format (₹)**.
- This changed values such as 69999 into ₹69,999.00.

Manufacturing Date:

- Applied the date format **DD-MM-YYYY**.

Formula: No formula was used; Excel cell formatting was applied.

7. Conditional Formatting

Price:

- Selected the Price column.
- Used **Conditional Formatting → Data Bars**.
- This visually shows the relative size of each price.

Category:

- Used Conditional Formatting with a rule to highlight cells where:

Category = Electronics

This makes the Electronics records easy to identify.

Formula: No formula was required for the basic conditional-formatting rules I used.

The original dataset is provided for reference before data cleaning and transformation.

